Beyond Unimodal Boundaries: Generative Recommendation with Multimodal Semantics

Jing Zhu [1] **Mingxuan Ju** [2] **Yozen Liu** [2] **Danai Koutra** [1]
Neil Shah [2] **Tong Zhao** [2]
[1] University of Michigan, Ann Arbor
[2] Snap
{jingzhuu, dkoutra}@umich.edu
{mju, yliu2, nshah, tzhao}@snapchat.com

arXiv:2503.23333v1 [cs.IR] 30 Mar 2025

Abstract

Generative recommendation (GR) has become a powerful paradigm in recommendation systems that implicitly links modality and semantics to item representation, in contrast to previous methods that relied on non-semantic item identifiers in autoregressive models. However, previous research has predominantly treated modalities in isolation, typically assuming item content is unimodal (usually text). We argue that this is a significant limitation given the rich, multimodal nature of real-world data and the potential sensitivity of GR models to modality choices and usage. Our work aims to explore the critical problem of *Multimodal Generative Recommendation (MGR)*, highlighting the importance of modality choices in GR frameworks. We reveal that GR models are particularly sensitive to different modalities and examine the challenges in achieving effective GR when multiple modalities are available. By evaluating design strategies for effectively leveraging multiple modalities, we identify key challenges and introduce MGR-LF++, an enhanced late fusion framework that employs contrastive modality alignment and special tokens to denote different modalities, achieving a performance improvement of over 20% compared to single-modality alternatives.

1 Introduction

Recommender systems help users discover content of interest and play an essential role in various domains such as videos (Covington et al., 2016), e-commerce (Schafer et al., 2001), and products (Geng et al., 2022). Recently, the *generative recommendation (GR)* which directly generates a ranked list of identifiers (IDs) for a given query using autoregressive models (e.g., large language models) has become increasingly popular (Sun et al., 2024; Rajput et al., 2023; Zheng et al., 2024; Deldjoo et al., 2024). As shown in Figure 1, at the core of GR models is the construction of *semantic IDs*, a sequence of discrete IDs which capture the semantic meaning of a document or item. Compared to existing approaches in sequential recommendation that typically assign a unique ID to each item, or traditional non-sequential recommendation approaches which often use random or frequency-based strategies for assigning IDs to items, semantic IDs aim to encapsulate the underlying characteristics or available modality information from items as an inductive prior to enable efficient retrieval and generalization to unseen items (Rajput et al., 2023; Lin et al., 2024; Wang et al., 2024; Zheng et al., 2024).

Despite the advancement of GR methods and prevalence of multimodal contents in recommendation systems (Liu et al., 2024), prior works have largely treated modality information exclusively, by inferring semantic IDs from unimodal (and often textual) content. However, GR is strongly grounded in the quality and availability of modality information, and treating it as unimodal raises questions about its effectiveness and sensitivity in multimodal environments, which are prevalent in real-world scenarios. Motivated by this limitation, we focus on the following fundamental question:

What are the key principles driving effective multimodal generative recommendation (MGR)?

To answer the question, we first explore the effectiveness of existing generative recommendation methods under multimodal environments. To enable their capability of taking multimodal data, we leverage early fusion and late fusion strategies (as illustrated in Figure 2). Early fusion employs a single multimodal encoder, such as ImageBind (Girdhar et al., 2023), to encode content from multiple modalities simultaneously, generating a unified set of semantic IDs that capture the semantics across all modalities. However, in practice, we found that one modality tends to dominate and there is significant information loss. In contrast, late fusion generates separate semantic IDs for each modality and combines them after the ID generation, but

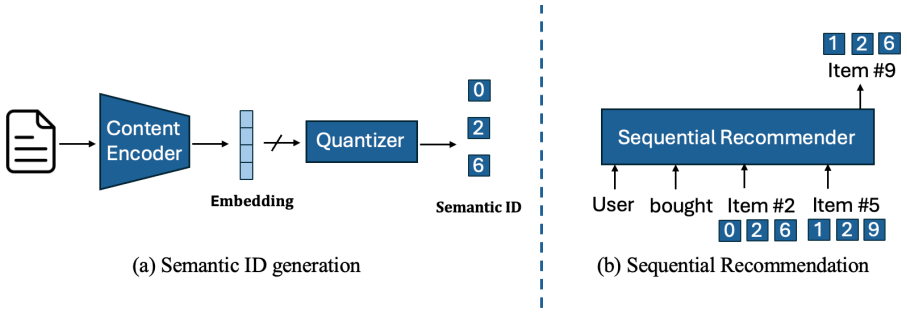


Figure 1: **Overview of a Typical Generative Recommendation Pipeline.** (a) Each item's semantic ID is generated as a list of discrete IDs based on its content. (b) These generated semantic IDs are then used to represent items within the sequential recommender.

our study highlights that it lacks a way to match corresponding semantic IDs across modalities.

Motivated by these observations, we first validate the importance of preserving the distinct information from each modality, which requires maintaining separate IDs, as done in late fusion. Moreover, to increase the effectiveness of the late fusion scheme, we propose two designs that alleviate the modality correspondence issue: (1) contrastive modality alignment training that directly matches corresponding semantic IDs across different modalities, and (2) special tokens that separate sequential IDs from different modalities.

Our contributions are summarized as follows:

- **Problem Formulation:** We present the first systematic study of multimodal generative recommendation or MGR, a practical but understudied research problem. We empirically evaluate early fusion and late fusion solutions, and identify the challenges that prevent existing GR solutions from working well under multimodal scenarios.
- **New Late Fusion Framework:** Based on our findings, we introduce MGR-LF++, an enhanced late fusion framework with two key designs that allow for maintaining the different modalities and aligning them for best performance.
- **Empirical Analysis:** Through extensive experiments with 6 baselines on 3 datasets, we show that carefully incorporating multimodal information into GR can significantly improve the performance (up to 20%) compared with its unimodal counterparts.

2 Preliminaries

We first present a typical generative recommendation pipeline, as illustrated in Figure 1. This pipeline comprises two distinct stages.

Stage 1: Semantic ID Generation. This stage involves encoding item content features using an off-the-shelf encoder to produce embedding vectors. These vectors are then quantized into a tuple of semantic codewords using an auto-encoder framework (e.g., RQ-VAE). The resulting tuple of codewords is referred to as the item's semantic ID.

Stage 2: Sequential Recommendation. Given a user's interaction history in the form of a sequence of semantic IDs (item_1, ..., item_n), the recommender system's task is to predict the next item, item_{n+1}, also in the form of a semantic ID.

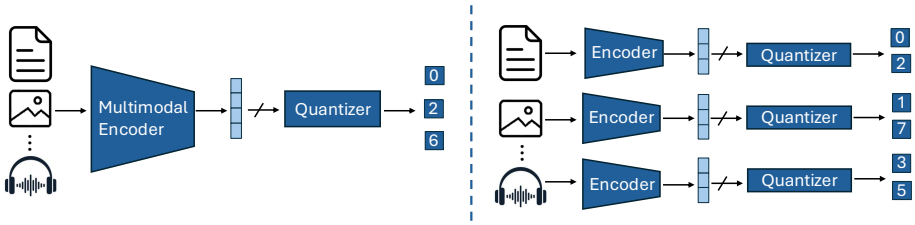
3 The MGR Problem

In this section, we begin by presenting the multimodal generative recommendation (MGR) problem, followed by a discussion of the unique challenges it poses and our proposed solutions to address these challenges.

Problem 1 (MGR) *Under a sequential recommendation problem setting, given all items with their associated multi-modal contents, how can we:*

- *Generate a list of discrete semantic IDs* $(c_{i_0}, ..., c_{i_{m-1}})$ *that effectively capture the semantics of multimodal contents, one for each item* i*?*
- *Design a sequential recommendation model that effectively integrates these semantic IDs for improved performance?*

To extend generative recommendation pipelines into multimodal environments, we begin by exploring two fundamental strategies: early fusion and late fusion. These approaches aim to effectively integrate multimodal content for generating semantic IDs, as illustrated in Figure 2.



(a) Early Fusion (b) Late Fusion

Figure 2: **Naive strategies for extending unimodal generative recommendation to multimodal scenarios.** Early fusion (MGR-EF) generates a unified list of semantic IDs capturing the semantics across all modalities. Late fusion (MGR-LF) generates separate semantic IDs for each modality and combines them after generation.

Table 1: **MGR-EF struggles to preserve semantics from all modalities, resulting in information loss.** We present Adjusted Mutual Information (AMI) scores between modalities. The AMI scores between text and multimodal consistently outweigh others across datasets, indicating that textual information dominates in the multimodal representations, while other modalities' contributions are diminished.

Datasets	Img vs. Txt	Img vs. Multimodal	Txt vs. Multimodal
Toys	0.2765	0.3892	0.4829
Beauty	0.2920	0.3937	0.5539
Sports	0.3976	0.5105	0.6351

Table 2: **MGR-EF produces similar prediction results to its unimodal counterparts, indicating potential information loss.** We present the percentage of overlapping predictions across modalities, revealing the frequency with which examples yield identical prediction results across different modality inputs.

Datasets	Img vs. Txt	Img vs. Multimodal	Txt vs. Multimodal
Toys	0.3287	0.3881	0.4612
Beauty	0.4724	0.4911	0.4866
Sports	0.2775	0.3675	0.4584

3.1 MGR-EF: Naive Early Fusion Framework

As shown in Figure 2(a), the early fusion strategy employs a single multimodal encoder to process content from multiple modalities concurrently. The resulting unified representation is then used to generate a single set of semantic IDs that encapsulate the semantics across all modalities. This approach ensures that inter-modal interactions are captured during the encoding phase, potentially leading to richer representations. However, early fusion may face challenges such as over-reliance on dominant modalities.

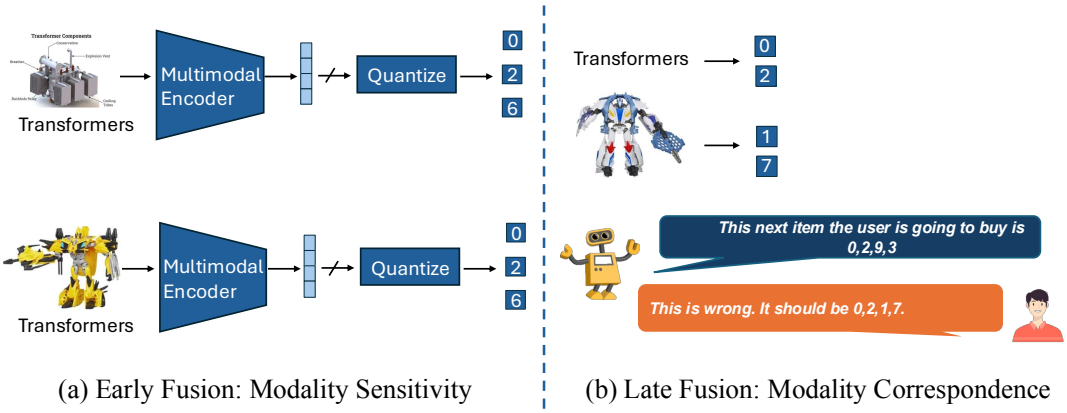
Finding 1: While the early fusion strategy is straightforward, it is susceptible to the modality sensitivity challenge, wherein the generated semantic IDs are dominated by one modality and fail to preserve the distinct semantics of multiple modalities. As illustrated in Figure 3(a), one modality may dominate the encoded information, resulting in identical semantic IDs for items that differ in other modalities.

To quantify this dominance, we employ two metrics: the Adjusted Mutual Information (AMI) score (defined in Appendix A.1) and the percentage of overlapping predictions in sequential recommendation, using text and image modalities as examples.

A higher AMI score indicates greater information overlap between generated semantic IDs, while a higher percentage of overlapping predictions suggests the sequential recommenders' inability to effectively utilize information from multiple modalities. As evidenced in Tables 1 and 2, the comparison between text and multimodal inputs yields the highest AMI score and percentage of overlapping hits. This finding indicates that the MGR-EF approach is compromised by modality sensitivity challenges, making it suboptimal for effective generative recommendation.

To address this issue, we propose **generating separate semantic IDs for each modality** before combining them, as implemented in late fusion, to mitigate the modality sensitivity challenge. Formally, given a set of modalities $M = m_0, m_1, ..., m_k$, we aim to generate a sequence of semantic IDs $C = [c_0, c_1, ..., c_n]$ where each c_i corresponds to a specific modality m_j. This approach facilitates better preservation of modality-specific information and potentially more balanced representation across different modalities in the final recommendation process.

The development of a more balanced early fusion-based approach capable of generating multimodal semantic IDs without information loss remains an avenue for future research.



(a) Early Fusion: Modality Sensitivity (b) Late Fusion: Modality Correspondence

Figure 3: **Key challenges in the MGR problem.** (a) Modality Sensitivity: Despite both items being transformers, they are distinct products. The semantic ID generated through early fusion fails to capture visual differences, resulting in information loss. (b) Modality Correspondence: The item's textual semantic ID is [0,2], with a corresponding visual semantic ID of [1,7]. While the sequential recommender correctly predicts the item as a transformer (by accurately predicting the textual semantic ID [0,2]), it fails to map this to the corresponding visual semantic ID [1,7].

Table 3: **MGR-LF fails correctly to generate the semantic IDs of multiple modalities sequentially.** We show Hits@5 of predicting semantic IDs of single modality versus multiple modalities.

Datasets	Text Only	Image Only	Multimodal
Toys	0.0598	0.0495	0.0482
Beauty	0.0505	0.0417	0.0336
Sports	0.0195	0.0193	0.0184

3.2 MGR-LF: Late Fusion Framework

As illustrated in Figure 2(b), late fusion generates separate semantic IDs for each modality independently using unimodal encoders. These modality-specific semantic IDs are subsequently combined through simple aggregation techniques (e.g., concatenation). This approach allows each modality to contribute independently before integration.

Although generating separate semantic IDs for each modality preserves modality-specific information, it introduces a significant challenge: modality correspondence. As evidenced in Table 3, there exists a substantial performance degradation when predicting the entire multimodal semantic ID sequence compared to predicting only a subset (Text Only or Image Only) of the sequence.

We hypothesize that the root cause of these challenges stems from the following findings:

Finding 2: The sequential recommender struggles to map generated semantic IDs for one modality to their corresponding IDs in other modalities. As depicted in Figure 3(b), while the sequential recommender correctly predicts the first two textual semantic IDs, it generates instead of the correct, indicating an inability to map the textual semantic ID sequence to its corresponding visual semantic ID sequence. Formally, when the sequential recommender generates a sequence of semantic IDs $C = [c_0, c_1, ..., c_n]$ for the next item, we observe that while it can correctly generate the first few IDs for one modality $c_{t_0}, .., c_{t_m}$, it fails to match the corresponding IDs $c_{v_0}, .., c_{v_m}$ in other modalities. This modality mismatch causes significant performance degradation, as the model would suffer from the false predictions of all modalities.

Finding 3: Multiple modalities introduce challenges in the conditional generation of semantic IDs. Sequential recommenders typically generate semantic IDs autoregressively, with each subsequent ID based on previously generated ones. This approach is effective for unimodal data, where each new ID acts as a residual for the preceding ones. However, it becomes problematic in multimodal contexts, as the semantic information of one modality often does not depend on that of others.

Consider a sequence of semantic IDs $C = [c_0, c_1, ..., c_n]$. When $c_{i-1} \in m_f$ and $c_i \in m_j$ (i.e., consecutive semantic IDs belong to different modalities), the conditional generation of c_i based on c_{i-1} becomes challenging. This is because c_i is not a residual for c_{i-1}, disrupting the conventional pattern of autoregressive generation.

3.3 MGR-LF++: Proposed Enhanced Late Fusion Framework

Motivated by our findings from analyzing the naive MGR-EF and MGR-LF frameworks, we focus on late fusion that generates semantic IDs per modality (Finding 1) and propose an enhanced LF framework:MGR-LF++ that incorporates two key designs that address the challenges identified in Findings 2 and 3.

Specifically, to address Finding 2, we propose a new training paradigm called **Contrastive Modality Alignment**. We first fine-tune the sequential recommender by explicitly tasking it to identify corresponding IDs for other modalities. After the sequential recommender gains knowledge about matching text and visual IDs, we then fine-tune it on the sequential recommendation task. Below are examples of the training paradigm, using text and image as example modalities:

Example

Image-to-Text Prediction: Source: Here we have an item with image id <img-1-40><img-2-80>.Please predict its corresponding text id.
Target: <txt-1-214><txt-2-250>.

Example

Text-to-Image Prediction: Source: Here we have an item with text id <txt-1-214><txt-2-250>.Please predict its corresponding image id.
Target: <img-1-40><img-2-80>.

Note that this contrastive alignment objective leverages only each item's semantic IDs, and no other information (e.g., user purchase history, item collaborative information) is used. Through this contrastive alignment, the model is able to match semantic IDs across different modalities.

To address Finding 3, we propose to incorporate **special tokens** $S = s_1, s_2, ..., s_{k-1}$ into the semantic ID sequences. Each s_i indicates a transition between modality $i-1$ and modality i. This results in a modified semantic ID sequence:

$$C = [c_0, .., c_i, s_1, c_{i+1}, .., s_{k-1}, .., c_n].$$

By explicitly marking modality transitions, this approach facilitates smoother integration of multimodal information while maintaining the integrity of the autoregressive generation process.

4 Experiments

In our empirical analysis, we aim to address the following research questions:

(RQ1) How effective is multimodal generative recommendation compared to unimodal generative recommendation?

(RQ2) What is the impact of our MGR-LF++ on multimodal generative retrieval, compared with MGR-EF and MGR-LF? How effective is each design of MGR-LF++?

(RQ3) How does MGR-LF++ perform across various ID lengths and codebook sizes?

4.1 Empirical Setup

Dataset Due to the lack of publicly available multimodal data, our experiments primarily focus on two modalities: text and images. However, our analysis is generalizable to other modalities. Following (Jin et al., 2024; Rajput et al., 2023), we conduct experiments on three domains from Amazon review dataset (Ni et al., 2019; Hou et al.), Toys, Beauty, and Sports. For sequential recommendations, we keep the users and items that have at least 5 interactions in their history. We treat the last interacted item by each user as the testing sample, the second to last second item as the validation sample, and the previous items as training samples. The text contents are the titles and descriptions of the products and the visual contents are the high-resolution raw images of the products. Items without raw images are removed. The statistics of the datasets can be found in Table 4. The data does not contain any information that names or uniquely identifies individual people or offensive content.

Table 4: Dataset statistics.

Dataset	# Users	# Items	# Rec history (train/val/test)
Toys	4502	4943	80% / 10% / 10%
Beauty	22334	12094	80% / 10% / 10%
Sports	35393	18287	80% / 10% / 10%

Evaluation Metrics For sequential recommendation, we report the three most commonly-used evaluation metrics: Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG), Hits@5 (Jin et al., 2024; Rajput et al., 2023). MRR averages the reciprocal rank of the first relevant item, emphasizing the importance of its position. NDCG evaluates ranking quality, weighting relevant items higher in the list using a logarithmic

discount. Hits@5 measures the proportion of times a relevant item appears within the top 5 recommendations. These metrics offer complementary insights: MRR focuses on the first relevant item's rank, NDCG evaluates the overall ranking quality, and Hits@5 measures the presence of relevant items within the top recommendations. Together, they provide a comprehensive assessment of the recommendation system's performance.

Implementation Details We use RQ-VAE as our indexer to generate discrete semantic IDs for each modality and T5 as our sequential recommender. For hyperparameter and computing resource details, please see Appendix A.2.

Baseline Methods We compare our method with two categories of baselines: **(1)** Various sequential recommendation models, including popular approaches such as SASREC (Kang and McAuley, 2018) and generative recommendation methods with semantic IDs like LETTER (Jin et al., 2024), HCindexer (Wang et al., 2024), and TIGER(BERT) (Rajput et al., 2023). **(2)** To validate the efficacy of multimodal generative retrieval, we compare our methods against corresponding unimodal counterparts using TIGER as the main backbone. These baselines are denoted as TIGER(CLIP-text) and TIGER(CLIP-image), where the parentheses indicate the feature encoders for semantic IDs. Notably, our MGR-EF, MGR-LF, and MGR-LF++ also utilize TIGER as the main backbone, with CLIP-text and CLIP-image serving as text and image feature encoders, respectively.

Proposed Methods We present MGR-EF and MGR-LF as two naive strategies for the MGR problem. Additionally, we report MGR-LF++ to demonstrate the results after incorporating our two key proposed designs: contrastive modality alignment and special tokens.

4.2 (RQ1) MGR vs. Unimodal GR

Table 5 presents the overall experimental results. Even the most basic multimodal strategies, MGR-EF and MGR-LF, demonstrate substantial performance improvements over unimodal counterparts such as TIGER(CLIP-text) by 7%. This underscores the value of extending existing unimodal generative retrieval frameworks to multimodal scenarios and highlights the significance of the MGR problem. Notably, LETTER exhibits poor performance in our experiments, attributable to the presence of unseen items without collaborative information in our datasets. As LETTER heavily relies on previously defined collaborative information, it fails to generate meaningful semantic IDs for these novel items.

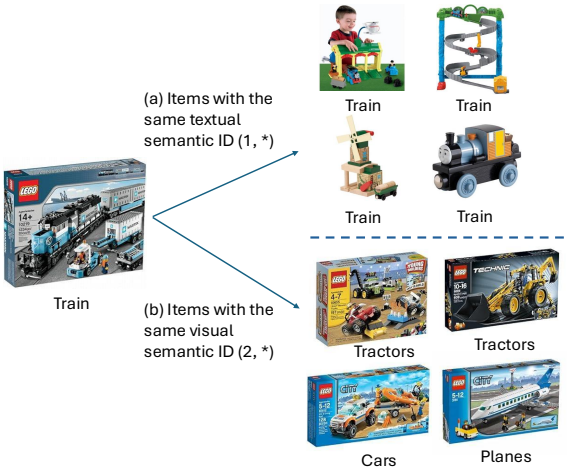


Figure 4: **Multimodality is needed: Semantic IDs of each modality captures different information.** Visualization of items with the same first semantic ID per modality wrt. the pivot example on Toys. (a) focuses on capturing items with similar textual semantics with various visual lookings while (b) focuses on capturing items that look similar to each other visually.

Furthermore, Figure 4 provides qualitative examples of items grouped by shared first textual or visual semantic IDs. Figure 4(a) demonstrates that textual semantic IDs capture items with similar textual meanings—in this case, all trains—while their visual representations vary significantly. Conversely, Figure 4(b) illustrates that items sharing visual semantic IDs appear visually similar—all LEGO toy boxes—despite representing different types of toys such as tractors, cars, and planes. When recommending similar items, both textual and visual similarity should be considered, as some users may prefer similar trains, while others may be interested in visually similar LEGO toys representing different vehicles.

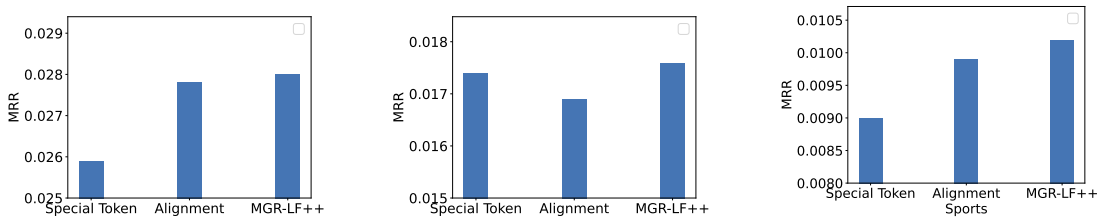
These findings demonstrate that leveraging semantic IDs from multiple modalities is essential for capturing diverse item relationships. This further underscores the importance of addressing the MGR problem to enhance generative recommendation in multimodal contexts.

4.3 (RQ2) MGR-LF++ vs. Naive Strategies

As evidenced in Table 5, MGR-LF++ outperforms the basic multimodal strategies, MGR-EF

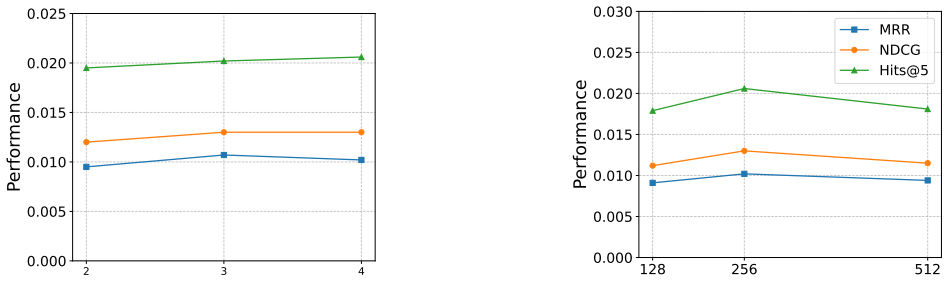
Table 5: **Performance of proposed MGR-LF++ and various baselines** across three sequential recommendation datasets. The best performance is highlighted in gray. MGR-LF++ achieves best performance across datasets.

Method	Toys			Beauty			Sports		
	MRR	NDCG	Hits@5	MRR	NDCG	Hits@5	MRR	NDCG	Hits@5
LETTER (Wang et al., 2024)	0.0169	0.0201	0.0300	0.0048	0.0059	0.0092	0.0018	0.0023	0.0036
HC-indexer (Jin et al., 2024)	0.0071	0.0088	0.0138	0.0074	0.0090	0.0138	0.0029	0.0036	0.0055
SASRec (Kang and McAuley, 2018)	0.0047	0.0056	0.0084	0.0084	0.0105	0.0167	0.0036	0.0042	0.0059
TIGER(BERT) (Rajput et al., 2023)	0.0173	0.0212	0.0331	0.0077	0.0101	0.0173	0.0050	0.0062	0.0098
TIGER(CLIP-text)	0.0217	0.0275	0.0451	0.0161	0.0199	0.0316	0.0103	0.0127	0.0203
TIGER(CLIP-image)	0.0198	0.0245	0.0389	0.0143	0.0177	0.0283	0.0062	0.0077	0.0121
MGR-EF (early fusion)	0.0231	0.0294	0.0486	0.0150	0.0187	0.0298	0.0086	0.0105	0.0163
MGR-LF (late fusion)	0.0220	0.0284	0.0482	0.0169	0.0210	0.0336	0.0091	0.0114	0.0184
MGR-LF++ (proposed)	0.0280	0.0349	0.0562	0.0176	0.0216	0.0334	0.0102	0.0130	0.0206



(a) Toys (b) Beauty (c) Sports

Figure 5: **Ablation Study on our two design choices across datasets.** Both of our proposed designs: special token for cross-modal indication and contrastive modality are important for the MGR problem.



(a) Semantic ID Length Per Modality (b) Codebook Size

Figure 6: Influence of (a) semantic ID length and (b) codebook size on MGR-LF++ on Sports. For larger datasets, it is important to have a longer semantic ID length. Setting codebook size to 256 gets the best performance.

and MGR-LF, across various datasets. MGR-LF++ achieves an average relative improvement of 20% compared to MGR-LF. This significant performance boost demonstrates the effectiveness of our proposed designs: contrastive alignment pre-training and special tokens. These results suggest that MGR-LF++ effectively addresses the challenges of modality sensitivity and modality correspondences previously discussed.

Figure 5 provides a more detailed analysis of how each design choice affects performance, with "alignment" referring to contrastive modality alignment. Both special tokens and contrastive modality alignment contribute to substantial performance improvements, with their combination yielding the best overall performance. For Toys and Sports, contrastive modality alignment is more important while for Beauty, adding special tokens between modalities is more important.

4.4 (RQ3) Ablation Studies

We investigate the impact of semantic ID length and codebook size on our framework's performance. Using the same experimental setup as in Sections 4.1, we conduct ablation studies on the Sports dataset, our largest dataset. The results are visualized in Figure 6.

Semantic ID Length As illustrated in Figure 6, longer semantic ID lengths generally yield better performance for larger datasets, despite slight vari-

ations across metrics. This improvement can be attributed to the richer representation provided by longer semantic IDs.

Codebook Size Our analysis, also depicted in Figure 6, reveals that a codebook size of 256 consistently produces the best overall performance across various metrics.

5 Related Work

Sequential Recommendation aims to predict a user's next interaction by modeling their historical behavior sequence. While traditional approaches relied on sequential pattern mining and Markov chains to capture interaction transitions (Fournier-Viger et al., 2017; Yap et al., 2012; Rendle et al., 2010), recent advances adapt transformer-based models to model long-range dependencies in user interaction sequences (Kang and McAuley, 2018; Zhu et al., 2024; Sun et al., 2019). However, these methods predominantly focus on collaborative signals and sequential patterns, often overlooking item content. Generative retrieval, which captures item content via semantic IDs, has emerged as a new paradigm for sequential recommendation. In this work, inspired by the recent advances of language models, we use autoregressive language model as our sequential recommenders.

Generative Recommendation, which directly generates a ranked list of identifiers (IDs) for a given query using LLMs, has emerged as a new paradigm for information retrieval (Sun et al., 2024; Rajput et al., 2023; Zheng et al., 2024; Deldjoo et al., 2024; Jin et al., 2024). At the core of these generative recommendation models is the construction of semantic IDs, a sequence of discrete ID numbers that captures the semantic meaning of a document. Current self-supervised semantic indexing methods generally follow a two-step process. In the first step, an off-the-shelf text encoder (Devlin et al., 2019) encodes input documents and generates embedding representations for them. In the second step, an auto-encoder framework (e.g., RQ-VAE) is typically leveraged to generate semantic IDs for each document. For the auto-encoder framework, the encoder learns discrete latent variables for input documents and the decoder reconstructs input from these discrete variables (Rajput et al., 2023; Sun et al., 2024; Jin et al., 2024; Wang et al., 2024). In this work, we extend the semantic ID generation pipeline to multimodal scenarios and uncover new challenges. Our work proposes new research questions and opens up new directions for generative recommendation.

Multimodal Representation Learning aims to learn effective representations that can capture information across different modalities, such as text, images, and audio. Transformer-based models have shown remarkable success in multimodal feature learning (Xu et al., 2023; Radford et al., 2021). These models typically learn transferable visual representations by leveraging corresponding natural language supervision. Models like FLAVA (Singh et al., 2022) and Perceiver (Jaegle et al., 2021) have demonstrated the effectiveness of jointly pre-training transformers on unpaired images and text, while CLIP has shown that contrastive objectives can effectively align representations from different modalities (Singh et al., 2022; Jaegle et al., 2021; Radford et al., 2021).For more details, we refer readers to the survey (Xu et al., 2023).

6 Conclusions

Our work highlights the critical role of modality choice in generative recommendation (GR) systems, challenging the unimodal assumptions prevalent in prior research. By introducing and exploring the problem of Multimodal Generative Recommendation (MGR), we highlight the untapped potential of leveraging multimodal content to enhance generative recommendation performance in real-world scenarios. Our findings demonstrate that GR models can be sensitive to modality selection and highlights the key challenges associated with MGR, when naive early/late fusion strategy is used: modality sensitivity and modality correspondence. To address these, we propose MGR-LF++, an enhanced late fusion framework that significantly improves performance by incorporating contrastive modality alignment and special tokens to denote modality transitions. This research not only advances our understanding of GR in multimodal contexts but also lays a foundation for future work aimed at optimizing the use of diverse modalities in generative recommendation systems.

Limitations

Our work showcases the promising potential of multimodal generative recommendation (MGR) and presents effective strategies, while also identifying several areas for further investigation.

First, our experiments reveal that early fusion is prone to modality sensitivity, where dominant modalities like text can overshadow others (e.g., images). Exploring a more balanced early fusion mechanism that can handle this challenge is worth investigating further.

Second, the datasets used in our experiments may not fully represent the diversity and complexity of real-world multimodal recommendation scenarios. For example, visual data in some datasets may be more informative than textual data or vice versa, which could bias the results toward certain modalities. Moreover, we primarily focused on text and image modalities due to the lack of publicly available datasets that have additional modalities (e.g., audio, video, tabular data). Future work should introduce and explore more diverse datasets to validate the generalizability of our findings.

References

Paul Covington, Jay Adams, and Emre Sargin. 2016. Deep neural networks for youtube recommendations. In Proceedings of the 10th ACM conference on recommender systems, pages 191–198.

Yashar Deldjoo, Zhankui He, Julian McAuley, Anton Korikov, Scott Sanner, Arnau Ramisa, René Vidal, Maheswaran Sathiamoorthy, Atoosa Kasirzadeh, and Silvia Milano. 2024. A review of modern recommender systems using generative models (gen-recsys). In Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pages 6448–6458.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Philippe Fournier-Viger, Jerry Chun-Wei Lin, Rage Uday Kiran, Yun Sing Koh, and Rincy Thomas. 2017. A survey of sequential pattern mining. Data Science and Pattern Recognition, 1(1):54–77.

Shijie Geng, Shuchang Liu, Zuohui Fu, Yingqiang Ge, and Yongfeng Zhang. 2022. Recommendation as language processing (rlp): A unified pretrain, personalized prompt & predict paradigm (p5). In Proceedings of the 16th ACM Conference on Recommender Systems, pages 299–315.

Rohit Girdhar, Alaaeldin El-Nouby, Zhuang Liu, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, and Ishan Misra. 2023. Imagebind: One embedding space to bind them all. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 15180–15190.

Y Hou, J Li, Z He, A Yan, X Chen, and JJ McAuley. Bridging language and items for retrieval and recommendation (2024). CoRR, abs/2403.03952.

Andrew Jaegle, Felix Gimeno, Andy Brock, Oriol Vinyals, Andrew Zisserman, and Joao Carreira. 2021. Perceiver: General perception with iterative attention. In International conference on machine learning, pages 4651–4664. PMLR.

Bowen Jin, Hansi Zeng, Guoyin Wang, Xiusi Chen, Tianxin Wei, Ruirui Li, Zhengyang Wang, Zheng Li, Yang Li, Hanqing Lu, et al. 2024. Language models as semantic indexers. ICML.

Wang-Cheng Kang and Julian McAuley. 2018. Self-attentive sequential recommendation. In 2018 IEEE international conference on data mining (ICDM), pages 197–206. IEEE.

Xinyu Lin, Wenjie Wang, Yongqi Li, Fuli Feng, See-Kiong Ng, and Tat-Seng Chua. 2024. Bridging items and language: A transition paradigm for large language model-based recommendation. In Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pages 1816–1826.

Qidong Liu, Jiaxi Hu, Yutian Xiao, Xiangyu Zhao, Jingtong Gao, Wanyu Wang, Qing Li, and Jiliang Tang. 2024. Multimodal recommender systems: A survey. ACM Computing Surveys, 57(2):1–17.

Jianmo Ni, Jiacheng Li, and Julian McAuley. 2019. Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP), pages 188–197.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In International conference on machine learning, pages 8748–8763. PMLR.

Shashank Rajput, Nikhil Mehta, Anima Singh, Raghunandan Hulikal Keshavan, Trung Vu, Lukasz Heldt, Lichan Hong, Yi Tay, Vinh Tran, Jonah Samost, et al. 2023. Recommender systems with generative retrieval. Advances in Neural Information Processing Systems, 36:10299–10315.

Steffen Rendle, Christoph Freudenthaler, and Lars Schmidt-Thieme. 2010. Factorizing personalized markov chains for next-basket recommendation. In Proceedings of the 19th international conference on World wide web, pages 811–820.

J Ben Schafer, Joseph A Konstan, and John Riedl. 2001. E-commerce recommendation applications. Data mining and knowledge discovery, 5:115–153.

Amanpreet Singh, Ronghang Hu, Vedanuj Goswami, Guillaume Couairon, Wojciech Galuba, Marcus Rohrbach, and Douwe Kiela. 2022. Flava: A foundational language and vision alignment model. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 15638–15650.

Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. 2019. Bert4rec: Sequential recommendation with bidirectional encoder representations from transformer. In Proceedings of the 28th ACM international conference on information and knowledge management, pages 1441–1450.

Weiwei Sun, Lingyong Yan, Zheng Chen, Shuaiqiang Wang, Haichao Zhu, Pengjie Ren, Zhumin Chen, Dawei Yin, Maarten Rijke, and Zhaochun Ren. 2024. Learning to tokenize for generative retrieval. Advances in Neural Information Processing Systems, 36.

Nguyen Xuan Vinh, Julien Epps, and James Bailey. 2009. Information theoretic measures for clusterings comparison: is a correction for chance necessary? In Proceedings of the 26th annual international conference on machine learning, pages 1073–1080.

Wenjie Wang, Honghui Bao, Xinyu Lin, Jizhi Zhang, Yongqi Li, Fuli Feng, See-Kiong Ng, and Tat-Seng Chua. 2024. Learnable item tokenization for generative recommendation. In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, pages 2400–2409.

Peng Xu, Xiatian Zhu, and David A Clifton. 2023. Multimodal learning with transformers: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence.

Ghim-Eng Yap, Xiao-Li Li, and Philip S Yu. 2012. Effective next-items recommendation via personalized sequential pattern mining. In Database Systems for Advanced Applications: 17th International Conference, DASFAA 2012, Busan, South Korea, April 15-19, 2012, Proceedings, Part II 17, pages 48–64. Springer.

Bowen Zheng, Yupeng Hou, Hongyu Lu, Yu Chen, Wayne Xin Zhao, Ming Chen, and Ji-Rong Wen. 2024. Adapting large language models by integrating collaborative semantics for recommendation. In 2024 IEEE 40th International Conference on Data Engineering (ICDE), pages 1435–1448. IEEE.

Yaochen Zhu, Liang Wu, Qi Guo, Liangjie Hong, and Jundong Li. 2024. Collaborative large language model for recommender systems. In Proceedings of the ACM on Web Conference 2024, pages 3162–3172.

A Appendix

A.1 AMI score

The Adjusted Mutual Information (AMI) score (Vinh et al., 2009) is a measure used in statistics and information theory to quantify the agreement between two clusters (in our experiments, the two clusters refer to ground truth category clusters and Semantic ID clusters) while correcting for chance. It is an adjustment of the Mutual Information (MI) score that accounts for the fact that MI is generally higher for clusters with a larger number of clusters, thus providing a normalized score that is more comparable across different clusters. Here we use AMI to calculate how much overlap the semantic IDs are across modalities. And since semantic IDs generated by RQ-VAE are hierarchical, we select the first semantic ID for each modality to compute the AMI score.

A.2 Implementation Details

For generative language model that are used as sequential recommender, we use T5-small and T5-base. When generating semantic IDs, the textual information (title & description) is encoded via the text encoder of CLIP and the visual information is encoder via the image encoder of CLIP to generate semantic embeddings for each modality. Take the embeddings as inputs, an rq-VAE indexer is used to generate semantic IDs for each item. The baselines use the same T5 checkpoint for fair comparison. Given the size of the datasets are different, we train 10,000 steps for toys, 15,000 steps for beauty and 20,000 steps for sports. For contrastive alignment pretraining, we first pretrain the corresponding steps using the contrastive objective and then use the pretrained checkpoint and train it directly on sequential recommedendation. We search learning rate in 1e-2, 1e-3, 1e-4. The maximum input text length is set to 1024 and all experiments are run on an 4 A100 40G machine. The number of beams for beam search is set to 20. Due to computation costs, all experiments are run once.